Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/30/2013
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kfk@wuphys.wustl.edu	**Fax:**	FY 314 935-6219
PI Organization Type:	UNIVERSITY	**Phone:**	314-935-6228
Organization Name:	Washington University		
PI Address 1:	Department of Physics, Campus Box 1105		
PI Address 2:	One Brookings Drive		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	10/01/2010	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: To be extended to 9/30/2015 per PI (Ed., 7/10/13)		
Key Personnel Changes/Previous PI:	Kenneth Kelton is U.S. Team Member on ESA-sponsored projects, “Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL),” and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AU19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled Vit106 and Vit106a glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of the liquid structure will be made as a function of the Ni concentration at the Advanced Photon Source using the ESL chamber at Washington University, which has been optimized for scattering studies. These will be correlated with the thermophysical property data obtained on ground and using the MSL-EML. Terrestrial maximum undercooling and induction time to recalescence data will be obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the MSL-EML undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids. The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering and a possible connection with liquid fragility. 3. Effect of diffusion on nucleation: Nucleation studies will be performed using the MSL-EML facility. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and the MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments will be compared with the results of ground-based ESL measurements. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state and ground-based ESL measurements of the maximum undercooling and induction time to recalescence in isothermal holds. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2 Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity will be made as a function of temperature. These will be coupled with x-ray diffraction measurements of the liquid structure factor (WP 2) to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalescece measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which has the potential to greatly improve the efficiency of materials development. However, this is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments planned under the support of this grant will allow these data to be obtained.
	NNX10AU19G is a continuation of an earlier grant (NNX07AK27G). NNX07AK27G, was focused primarily on ground-based studies in anticipation of a flight opportunity for a planned ISS experiment (QUASI); the focus of that grant was to support ground-based measurements of structures and thermophysical properties of supercooled high-temperature liquids that would support that flight experiment. QUASI now has flight opportunities on an ESA facility, working in collaboration with two European teams (THERMOLAB and ICOPROSOL). The focus of NNX10AU19G is to continue to support the ground-based structural, density, surface tension and viscosity studies of supercooled liquids, and to support ISS-flight-based studies of specific heat, emissivity, thermal conductivity, viscosity, and nucleation studies in a micro-gravity environment. Working with the THERMOLAB team, QUASI has one sample in Batch 1 (Vit 106a); QUASI is the team leader for another sample in the same batch, Vit 106. Additionally, QUASI has formally stated interest in several others samples in Batch 1. The definition of the Batch 2 samples is well underway. As part of the THERMOLAB team, QUASI will be allowed one sample in that batch; as part of ICOPROSOL, it will be allowed another. Again, QUASI has formally stated interest in many other samples in that batch and will work with the lead teams to design the experimental study and participate in the measurements. In addition, QUASI was a partner on two recently successful ESA proposals (THERMOLAB and ICOPROSOL), which were chosen for flight opportunities in Batches 3 and 4, and possibly in subsequent batches.

Task Progress:

This past year, NNX10AU19G and its predecessor (NNX07AK27G) grants have provided partial support for the PI, a Research Scientist, and two graduate students. The key activities and results of ground-based studies over this period are listed below.

Activities

The key activities over the past year are listed below:

1. Support of two Internal Working Group meetings: (i) November 15-16, 2012 at Astrium, Friedrichshafen, Germany, and (ii) March 8-9, 2013 in San Antonio, TX following the TMS annual meeting.

2. Prepared and delivered flight samples for the upcoming ISS experiment.

3. Supported a visit to MUSC from July 24 – August 5, 2013 to determine sample parameters for the upcoming ISS experiments.

4. Ground-based measurements of the volume, thermal expansion, and specific heat of the Vit 106 and Vit 106a liquids as a function of temperature using WU-BESL.

5. Measurements of the viscosity of Cu-Zr liquids using WU-BESL, which demonstrate the correlation between our measured thermal expansion coefficients and liquid fragility.

6. Measurements of the viscosity of supercooled metallic liquids using WU-BESL and correlation with liquid fragility.

7. Measurements of the surface tension of supercooled metallic liquids using WU-BESL. These demonstrate that sample rotation must be taken into account for a quantitative analysis of such data.

8. Measurements of the evaporation rates for samples for future study on the ISS (Batch #2).

9. Synchrotron X-ray diffraction measurements of liquids of interest for the current and future batches on the ISS.

10. Identification of the general trend that the first coordination shell in supercooled liquids shrinks with increasing temperature and the development of an explanation for this anomalous behavior.

11. Completed studies of the spherical absorption correction for X-ray diffraction data from levitated liquids and published the results.

Results of Ground-Based Activities

The experimental ground-based activities are aimed at providing complimentary data to the results that will be obtained on the ISS. The ground-based data helps to better define the experimental investigations on the ISS and provides a framework for the interpretation of the data obtained on orbit. It also contributes to a deeper understanding of the fundamental science that is central to these studies. Ground based activities over the past year have focused on measurements of the physical properties (density, surface tension, viscosity, evaporation rates, structural evolution) of supercooled liquids in a containerless environment using the electrostatic levitation facility located at Washington University (WU-BESL). The samples chosen for study are largely ones that have been selected, or are potential selections, for flight studies on the ISS. Selected results are briefly reviewed here.

1. Method for temperature calibration for optical pyrometry in containerless studies – The measurement and interpretation of physical property data obtained from samples in a containerless environment require accurate and precise measurements of the sample temperature. This is typically obtained using optical pyrometers, which must be calibrated. While it should be possible to make this calibration by matching known features in the chemical phase diagrams, these are often unavailable or inaccurate for the high melting temperature metallic alloys of interest. We have, therefore, developed and tested a technique that matches thermal features measured in the WU-BESL with corresponding features measured using a differential scanning calorimetry. This technique will also be applied to the containerless studies that will be made on the ISS.

2. Cu-Zr liquids – correlation with fragility – A key concept used to classify all liquids is that of fragility. Fragility has played a central role in helping to understand the kinetic behavior of a liquid, such as its viscosity, and how that liquid becomes a glass. Liquids are classified as “strong” when the logarithm of the viscosity as a function of inverse temperature has a constant slope (activation energy). They are “fragile” when the slope of this plot becomes increasingly larger on approaching the glass transition temperature (defined as the temperature where the viscosity attains the value of 10 to the power 12 in units of Pa.s). While fragility is typically assessed from measurements of the viscosity, it should also be evident in structural properties. We have demonstrated that this is the case by making simultaneous density and viscosity measurements for a series of Cu-Zr liquids of different chemical composition. Peaks in the activation energies and thermal expansivities at temperatures near the melting temperature are also noted at the compositions in these liquids that most easily form glasses, supporting a strong connection between fragility and glass formability. Fragility is typically measured for liquids near the glass transition temperature, which is not possible in most cases. Most liquids crystallize before this temperature can be reached. Our results are the first to clearly demonstrate that fragility can be assessed from measurements in the high temperature liquid, avoiding the problems of crystallization and opening the door for more extensive studies on a range of systems. The results of these measurements can be directly compared with the results obtained from molecular dynamics (MD) studies. This is important because the cooling rates that are possible in MD simulations are orders of magnitude larger than are experimentally accessible, making predictions of liquid structures and dynamics near the glass transition temperature meaningless.

3. Studies of supercooled liquids of different fragility – Given the success in the Cu-Zr liquids, we are expanding our studies to other liquids of interest for the ISS investigations. In Cu55-xAlxZr45 liquids, for example, we again find that the fragility determined from viscosity measurements of the high temperature liquid correlates with glass formability. A comparison between the fragilities measured in many high temperature liquids with the rate of structural evolution determined from X-ray synchrotron studies of the supercooled liquid has demonstrated directly the expected connection between dynamical fragility and liquid structure. This is the first time that this has been possible.

4. Thermal evaporation studies of ISS samples for Batch 2 – The definition of samples for the Batch 2 studies on the ISS is nearing completion. QUASI is using the WU-BESL to measure the evaporation rates of seven of these samples (Cu46Zr54, Cu50Zr50, LN-601, Zr64Ni36, Ni59.5Nb40.5, Ti39.5Zr39.5Ni21, and Cu47Zr47Al6). These data allow a computation of the coil deposition and dust generation resulting from processing, which is important to ensure the

	lifetime of the facility and the health of the astronauts on the ISS. Additionally, they provide new data for assessing thermodynamic parameters. The evaporation rates of all samples are lower than a rule of mixtures estimate using the pure elements (Langmuir equation). The deviation from the Langmuir equation becomes larger at lower temperatures. These differences reflect the attractive bonding between the elements. The increasing deviation at lower temperatures reflects the smaller contribution from the entropy of mixing. In addition, four articles are in press in the journal, International Journal of Thermophysics, and publication is expected in December 2013: Bendert JC, Pueblo, CE, Veligati S, Mauro, NA and Kelton KF "Temperature calibration for optical pyrometry in containerless system using differential scanning calorimetry - Application to Cu100-xZrx (X=45,50)" Int. J. Thermophysics (in press, 2013), Dec-2013 Bendert JC and Kelton KF "Containerless measurements of density and viscosity for a Cu48Zr52 liquid" Int. J. Thermophysics (in press, 2013), Dec-2013 Bradshaw RC, Hyers RW, Rogers JR, Rathz TJ, Lee GW, Gangopadhyay AK and Kelton KF "Non-contact techniques for measurement of density: data for a quasicrystal forming Ti-Zr-Ni alloy" Int. J. Thermophysics (in press, 2013), Dec-2013 Blodgett M, Gangopadhyay AK and Kelton KF "Thermal evaporation measurements for quasicrystal (TiZrNi) and glass forming (Vit 106 and 106a) liquids" Int. J. Thermophysics (in press, 2013), Dec-2013
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